Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce, and Science College, Ahmednagar (Autonomous)

(Affiliated to Savitribai Phule Pune University, Pune)

National Education Policy (NEP)
Choice Based Credit System (CBCS)

Programme Skeleton and Syllabus of
B. Sc. (Computer Science)-III (Major)

Implemented from

Academic Year 2025-26

Credit Distribution: B.Sc. Computer Science (Major) including Minor and OE and other courses.

	Type of Courses	III Yr	IV Yrs (Honours)	IV Yrs Research
Major Computer Science	Discipline-Specific Courses (DSC)	46	74	66
	Discipline Specific Elective (DSE)	08	16	16
	Skill Enhancement Courses (SEC)	06	06	06
	Vocational Skill Courses (VSC)	08	08	08
	On-Job Training (OJT)	04	08	04
	Field Project (FP)	04	04	04
	Community Engagement and Service (CEP)	02	02	02
	Research project	00	00	12
	Research Methodology	00	04	04
	Indian Knowledge System	02	02	02
	Total (I, II and III Year)	80	124	124
Minor	Minor	20	20	20
Other Courses	Open Elective (OE)/ Multidisciplinary Courses	12	12	12
	Co-Curricular Courses	08	08	08
	Ability Enhancement Courses	08	08	08
	Value Education Courses	04	04	04
	Total	132	176	176

B. Sc. Programme Framework: Credit Distribution

Year	Semester	Level	Major											Minor	OE	CC	AEC	VEC	Total
			DSC		DSE		SEC		VSC		FP/ OJT /IN/CEP		IKS						
			T	P	T	P	T	P	T	P	T	P		T/P	-	-	-	-	-
I	I	4.5	4	2	-	-	-	2	-	-	-	-	2	03	3	2	2	2	22
I	II	4.5	6	-	-	-		2	-	2	-	-		03	3	2	2	2	22
Exit Option: Award of UG Certificate in Major with 44 credits and an additional 4 credit core NSQF course /Internship or Continue with Major and Minor																			
II	III	5.0	6	2	-	-		2	-	-	-	2		03	3	2	2	-	22
II	IV	5.0	6	2	-	-		-	-	2	-	2		03	3	2	2	-	22
Exit Option: Award of UG Diploma in Major with 88 credits and an additional 4 credit core NSQF course /Internship or Continue with major and minor																			
III	V	5.5	8	2	2	2	-	-	-	2		2		04	-	-	-	-	-	22
III	VI	5.5	6	2	2	2	-	-	-	2		4		04	-	-	-	-	-	22
Exit Option: Award of UG Degree in Major and Minor with 132 credits or continue with Major for a 4-year Degree																			

IV	VII	6.0	8	6	2	2	RM-4		-	-	-	-		-	-	-	-	-	-	-	22
IV	VIII	6.0	8	6	2	2	-	-	-	-	-	4		-	-	-	-	-	-	-	22
Four Year UG Degree(Honours) with Major and Minor with 176 credits																					
IV	VII	6.0	6	4	2	2	RM-4		-	-	-	4		-	-	-	-	-	-	-	22
IV	VIII	6.0	6	4	2	2	-	-	0	-	-	8		-	-	-	-	-	-	-	22
Four Year UG Degree (Honours with Research) with Major and Minor with 176 credits																					

B. Sc. Programme Framework: Course Distribution

Year	Semester	Level	Major											Minor		OE	CC	AEC	VEC	Total
			DSC		DSE		SEC		VSC		FP/ OJT /IN/CEP		IKS							
-	-	-	T	P	T	P	T	P	T	P	T	P		T	P	-	-	-	-	-
I	I	4.5	2	1	-	-	-	1	-	-	-	-	1	1		1	1	1	1	10
I	II	4.5	2	-	-	-		1	-	1	-	-		1		1	1	1	1	09
Exit Option: Award of UG Certificate in Major with 44 credits and an additional 4 credit core NSQF course /Internship or Continue with major and minor																				
II	III	5.0	2	1	-	-		1	-	-	-	1		1		1	1	1	-	09
II	IV	5.0	2	1	-	-		-	-	1	-	1		1		1	1	1	-	09
Exit Option: Award of UG Diploma in Major with 88 credits and an \additional 4 credit core NSQF course /Internship or Continue with major and minor																				
III	V	5.5	2	1	1	1	-	-	-	1		1		1		-	-	-	-	08
III	VI	5.5	2	1	1	1	-	-	-	1		1		1		-	-	-	-	08
Exit Option: Award of UG Degree in Major and Minor with 132 credits or continue with Major for a 4-year Degree																				
IV	VII	6.0	3	3	1	1	0	1	-	-	-	-		-	-	-	-	-	-	09
IV	VIII	6.0	3	3	1	1	-	-	-	-	-	1		-	-	-	-	-	-	09
Four Year UG Degree(Honours) with Major and Minor with 176 credits																				
IV	VII	6.0	2	2	1	1	0	1	-	-	-	1		-	-	-	-	-	-	08
IV	VIII	6.0	2	2	1	1	-	-	-	-	-	1		-	-	-	-	-	-	07
Four Year UG Degree (Honours with Research) with Major and Minor with 176 credits																				

Programme Framework (Course Distribution): B.Sc. (Computer Science) (Major)

Year	Semester	Level	Major											Total	
			DSC		DSE		SEC		VSC		FP/OJT /IN/CEP/PR		IKS		
			T	P	T	P	T	P	T	P	T	P	T	T	P/PR
I	I	4.5	2	1	-	-	-	1	-	-	-	-	01	03	02
I	II	4.5	2	-	-	-		1	-	1	-	-		02	02
II	III	5.0	2	1	-	-		1	-	-	-	1		02	03
II	IV	5.0	2	1	-	-		-	-	1	-	1		02	03
III	V	5.5	2	1	1	1	-	-	-	1		1		03	04
III	VI	5.5	2	1	1	1	-	-	-	1		1		03	04
B.Sc. Honours															
IV	VII	6.0	3	3	1	1	RM-1		-	-	-	-		05	04
IV	VIII	6.0	3	3	1	1	-	-	-	-	-	1		04	05
B.Sc. Honours with Research															
IV	VII	6.0	2	2	1	1	RM-1		-	-	-	1		04	04
IV	VIII	6.0	2	2	1	1	-	-	-	-	-	1		03	04

Programme Framework (Credit Distribution): B.Sc. (Computer Science) (Major)

Year	Semester	Level	Major											Total
			DSC		DSE		SEC		VSC		FP/OJT /IN/CEP/RP		IKS	
			T	P	T	P	T	P	T	P	T	P	**T**	
I	I	4.5	4	2	-	-	-	2	-	-	-	-	02	10
I	II	4.5	6	-	-	-		2	-	2	-	-		10
II	III	5.0	6	2	-	-		2	-	-	-	2		12
II	IV	5.0	6	2	-	-		-	-	2	-	2		12
III	V	5.5	8	2	2	2	-	-	-	2		2		18
III	VI	5.5	6	2	2	2	-	-	-	2		4		18
IV	VII	6.0	8	6	2	2	RM-4		-	-	-	-		22
IV	VIII	6.0	8	6	2	2	-	-	-	-	-	4		22
IV	VII	6.0	6	4	2	2	RM-4	-	-	-	-	4		22
IV	VIII	6.0	6	4	2	2	-	-	-	-	-	8		22

Programme Framework (Courses and Credits): B.Sc. (Computer Science) (Major)

Sr. No.	Year	Semester	Level	Course Type	Course Code	Title	Credits
1.	I	I	4.5	DSC-1	BS-CS111T	Problem Solving using Computer and 'C' Programming	02
2.	I	I	4.5	DSC-2	BS- CS112T	Database Management Systems	02
3.	I	I	4.5	DSC-3	BS- CS113P	Lab Course on C Programming & DBMS	**02**
4.	I	I	4.5	SEC-1	BS- CS114T	Analog Electronics	02
5.	I	I	4.5	IKS-1	BS- CS115T	Science and Technology in Ancient India	**02**
6.	I	II	4.5	DSC-4	BS-CS121T	Advanced 'C' Programming	**03**
7.	I	II	4.5	DSC-5	BS- CS122T	Relational Database Management Systems	**03**
8.	I	II	4.5	SEC-2	BS- CS123P	Lab Course on Advanced C Programming & RDBMS	02
9.	I	II	4.5	VSC-1	BS- CS124 T	Digital Electronics	02
10.	II	III	5.0	DSC-6	BS- CS 231T	Data Structures and Algorithms	03
11.	II	III	5.0	DSC-7	BS-CS232T	NoSQL	03
12.	II	III	5.0	DSC-8	BS-CS233P	Lab Course on Data Structure and Algorithms	02
13.	II	III	5.0	SEC-3	BS-CS234P	Lab Course on NoSQL	02
14.	II	III	5.0	FP-01	BS-CS235P	Field Project using Software Engineering Concepts	02
15.	II	IV	5.0	DSC-9	BS-CS241T	Object Oriented Concepts using C++	03
16.	II	IV	5.0	DSC-10	BS-CS242T	Core Python	03
17.	II	IV	5.0	DSC-11	BS-CS243P	Lab Course on Object Oriented Concepts using C++	02
18.	II	IV	5.0	VSC-2	BS-CS244P	Lab Course on Core Python	02
19.	II	IV	5.0	CEP-01	BS-CS245P	Project	02
20.	III	V	5.5	DSC-12	BS- CS351T	Operating System	02

21.	III	V	5.5	DSC-13	BS- CS352T	Internet Programming	02
22.	III	V	5.5	DSC-14	BS- CS353T	Professional Practices in Python	02
23.	III	V	5.5	DSC-15	BS- CS354T	Object Oriented Software Engineering	02
24.	III	V	5.5	DSC-16	BS- CS355P	Lab Course Based on BS-CS352Tand BS- CS353T	02
25.	III	V	5.5	DSE-01	BS-CS356T(A)	Core Java	02
OR							
	III	V	5.5	DSE-01	BS-CS356T(B)	Data Analytics	
26.	III	V	5.5	DSE-02	BS-CS357P(A)	Lab Course Based on BS-CS356T(A)	02
OR							
	III	V	5.5	DSE-02	BS-CS357P(B)	Lab Course Based on BS-CS356T(B)	02
27.	III	V	5.5	VSC-03	BS- CS358T	Theoretical computer Science	02
28.	III	V	5.5	FP-02	BS- CS359P	Field Project	02
29.	III	VI	5.5	DSC-17	BS- CS360T	React Js	02
30.	III	VI	5.5	DSC-18	BS- CS361T	Compiler Construction	02
31.	III	VI	5.5	DSC-19	BS- CS362T	Android Programming	02
32.	III	VI	5.5	DSC-20	BS- CS363P	Lab Course Based on BS-CS360T	02
33.	III	VI	5.5	DSE-03	BS-CS364T(A)	Advanced Java	02
OR							
	III	VI	5.5	DSE-03	BS-CS364T(B)	Soft Computing	02
34.	III	VI	5.5	DSE-04	BS- CS365P (A)	Lab Course Based on BS-CS 364T(A)	02
OR							
	III	VI	5.5	DSE-04	BS- CS365P (B)	Lab Course Based on BS-CS 364T(B)	02
35.	III	VI	5.5	VSC-04	BS- CS366P	Lab Course Based on BS-CS 362T	02
36.	III	VI	5.5	OJT-01	BS- CS367P	On Job Training	04

B.Sc. (Computer Science) (Major with Honours)

1.	IV	VII	6.0	DSC-18	BS-CH471T	KLM	03
2.	IV	VII	6.0	DSC-19	BS-CH472T	NOP	03
3.	IV	VII	6.0	DSC-20	BS-CH473Y	QRS	02
4.	IV	VII	6.0	DSC-21	BS-CH474P	TUV	02
5.	IV	VII	6.0	DSC-22	BS-CH475P	WXY	02
6.	IV	VII	6.0	DSC-23	BS-CH476P	YXZ	02
7.	IV	VII	6.0	DSE-05	BS-CH477T	YXZ	02
8.	IV	VII	6.0	DSE-06	BS-CH478P	YXZ	02
9.	IV	VII	6.0	RM-01	BS-CH479T/P	ZAB	04
10.	IV	VIII	6.0	DSC-24	BS-CH481T	CDE	03
11.	IV	VIII	6.0	DSC-25	BS-CH482T	FGH	03
12.	IV	VIII	6.0	DSC-26	BS-CH482T	YGK	02
13.	IV	VIII	6.0	DSC-27	BS-CH482P	MYK	02
14.	IV	VIII	6.0	DSC-28	BS-CH482P	VYK	02
15.	IV	VIII	6.0	DSC-29	BS-CH482P	RYK	02
16.	IV	VIII	6.0	DSE-07	BS-CH485T	ABC	02
17.	IV	VIII	6.0	DSE-08	BS-CH486P	CBA	02
18.	IV	VIII	6.0	OJT-02	BS-CH487P	RST	04

B.Sc. Computer Science (Major Honours with Research)

34.	IV	VII	6.0	DSC-20	BS-CH471T	KLM	03
35.	IV	VII	6.0	DSC-21	BS-CH472T	NOP	03
36.	IV	VII	6.0	DSC-22	BS-CH473P	TUV	02
37.	IV	VII	6.0	DSC-23	BS-CH474P	WXY	02
38.	IV	VII	6.0	DSE-05	BS-CH475T	WXY	02
39.	IV	VII	6.0	DSE-06	BS-CH476P	YXZ	02
40.	IV	VII	6.0	RM-01	BS-CH477T/P	ZAB	04
41.	IV	VII	6.0	RP-01	BS-CH488P	AAA	04
42.	IV	VIII	6.0	DSC-20	BS-CH481	KLM	03
43.	IV	VIII	6.0	DSC-21	BS-CH482	NOP	03
44.	IV	VIII	6.0	DSC-22	BS-CH483	TUV	02
45.	IV	VIII	6.0	DSC-23	BS-CH484	WXY	02
46.	IV	VIII	6.0	DSE-07	BS-CH485	WXY	02
47.	IV	VIII	6.0	DSE-08	BS-CH486	YXZ	02
48.	IV	VIII	6.0	RP-02	BS-CH487	AAA	08

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar (Autonomous)

Board of Studies in Center for advanced studies in Applied Sciences.

Sr. No.	Name	Designation
1.	Prof. A.D. Gangarde	Chairman
2.	Prof. P.U. Patil	Member
3.	Prof. M.B. Bhingare	Member
4.	Prof. M.B. Gobare	Member
5.	Prof. B.M..Danave	Member
6.	Prof. D.V.Jagdale	Member
7.	Prof. Vaibhav Ekshinge	Member
8.	Dr. M.N.Shelar	Hon.Vice-Chancellor Nominee
9.	Dr. Shankar Mali	Academic Council Nominee
10.	Dr. Chandrashekhar Patil	Academic Council Nominee
11.	Mr. U.C.Temkar	Industrial Expert
12.	Mr. Prajwal Bhalsingh	Post Graduate Meritorious
13	Prof. D.S. Shelar	Co-Opt (Electronics)
14	Prof. S.A. Tarate	Co-Opt (Mathematics)
15	Dr. A.A. Kulkarni	Co-Opt (Statistics)
16	Mr. Vineet Chhajed	Invitee

Prologue/ Introduction of the programme:

Education is the key to development of any society. Role of higher education is crucial for securing right kind of employment and also to pursue further studies in best available world class institutes elsewhere within and outside India. Quality education in general and higher education in particular deserves high priority to enable the young and future generation of students to acquire skill, training and knowledge in order to enhance their thinking, creativity, comprehension and application abilities and prepare them to compete, succeed and excel globally. Sustained initiatives are required to reform the present higher education system for improving and upgrading the academic resources and learning environments by raising the quality of teaching and standards of achievements in learning outcomes across all undergraduate programs in science, humanities, commerce and professional streams of higher education including computer science. B.Sc. (Computer Science) has been evolving as an important branch of science and engineering throughout the world in last couple of decades and it has carved out a space for itself like any other disciplines of basic science and engineering. Computer science is a discipline that spans theory and practice and it requires thinking both in abstract terms and in concrete terms. Computer science has a wide range of specialties. These include Computer Architecture, Software Systems, Graphics, Artificial Intelligence, Computational Science, and Software Engineering. Drawing from a common core of computer science knowledge, each specialty area focuses on specific challenges. Computer Science is practiced by mathematicians, scientists and engineers. Mathematics, the origins of Computer Science, provides reason and logic. Computer Science education at undergraduate level (+3) will result in earning Bachelor of Science (BSc) degree in Computer Science. The coursework required to earn a BSc is equally weighted in mathematics and science. B.Sc. with Computer Science are aimed at undergraduate level training facilitating multiple career paths. Students so graduated, can take up postgraduate programmes in Computer Science and can be employable at IT industries. There are several employment opportunities and after successful completion of an undergraduate programme in Computer Science, graduating students can fetch employment directly in companies as Web Developer, Software Engineer, Network Administrator, Data Scientist, or AI/ML personnel.

2**. Programme Outcomes (POs)**

1. Demonstrate the aptitude of Computer Programming and Computer based problem solving skills.
2. Display the knowledge of appropriate theory, practices and tools for the specification, design, implementation.
3. Ability to link knowledge of Computer Science with other two auxiliary disciplines of study.
4. Display ethical code of conduct in usage of Internet and Cyber systems.
5. Ability to pursue higher studies of specialization and to take up technical employment.
6. Ability to formulate, to model, to design solutions, procedure and to use software tools to solve real world problems and evaluate.
7. Ability to operate, manage, deploy, configure computer network, hardware, software operation of an organization.
8. Ability to present result using different presentation tools.
9. Ability to appreciate emerging technologies and tools.
10. Apply standard Software Engineering practices and strategies in real-time software project development.
11. Design and develop computer programs/computer -based systems in the areas related to algorithms, networking, web design, cloud computing, IoT and data analytics.
12. The ability to work independently on a substantial software project and as an effective team member.

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc. (Computer Science)-III (Major)

Title of the Course: Operating System								
Year: III				**Semester: V**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-12	BS-CS351T	02	00	02	30	15	35	50

Learning Objectives:

- To provide a deep understanding of operating system concepts, including process management,
- memory management, file systems, and security.
- To familiarize students with real-world OS implementations (e.g., Linux, Windows).
- To develop practical skills in OS-related tasks, such as shell scripting, process scheduling, and
- system monitoring.
- To prepare students for IT industry roles by emphasizing troubleshooting, performance
- optimization, and system administration.

Course Outcomes (Cos)

On completion of the course, student will be able to–

- concept of operating system and its principle
- Processes and Thread Scheduling by operating system
- Synchronization in process and threads by operating system
- Memory management by operating system using with the help of various schemes

Detailed Syllabus:

UNIT I: Introduction to Operating Systems (6)

- OS Definition, Objectives, Functions | Types: Batch, Multiprogramming
- Time-sharing, Real-time, Distributed, Mobile OS | Kernel Types
- OS Structures | System Calls & APIs
- OS Services | Linux Basics
- Windows Basics | Shell Scripting Intro Module

UNIT II: Process Management (10)

- Process States, PCB | Context Switching
- Process Creation/Termination (fork(), exec())
- Threads vs. Processes | Multithreading Model
- Scheduling: FCFS, SJF, Round Robin
- Priority, Multilevel Queue | Linux/Windows Schedulers

- IPC: Shared Memory | Message Passing
- Synchronization: Mutex, Semaphores
- Deadlocks: Prevention, Avoidance, Detection

UNIT III: Memory Management **(6)**

- Memory Hierarchy | Contiguous Allocation
- Paging | Segmentation
- Page Tables, TLB | Demand Paging
- Page Replacement: FIFO, LRU, Optimal
- Thrashing | Linux/Windows Memory Management

UNIT IV: File Systems and Storage Management **(8)**

- File Attributes, Operations | Directory Structures
- Allocation Methods | Free Space Management
- Disk Structure | Scheduling: FCFS, SSTF
- SCAN, C-SCAN, LOOK
- Linux ext4 | Windows NTFS

UNIT V: Security and Protection **(5)**

- Authentication | Authorization Models
- Encryption | Threats (Malware, DDoS)
- ACLs | Capabilities
- Linux/Windows Permissions
- OS Hardening

Suggested Textbooks and Resources

1. Textbooks:
   Operating System Concepts by Abraham Silberschatz, Peter B. Galvin, and Greg Gagne.
   Modern Operating Systems by Andrew S. Tanenbaum.

2. Online Resources:
   Linux man pages and documentation.
   Microsoft Windows documentation.
   Online courses (e.g., Coursera, edX).

3. Tools:
   VirtualBox/VMware for OS virtualization.
   Linux terminal and Windows PowerShell.

Practical Assignments for the Operating Systems course

However due to NEP structure it is not possible to incorporate practical course for this subject as a compulsory practical credit course. To get aware with practical exposure and environment following is the list of practical's expected to be completed by the students as home assignment. This assignment will help students to understand the concepts of the operating systems.

1. Shell Scripting Assignments

Assignment 1: Basic Shell Scripting

- Task: Write a Bash script to automate the following tasks:

1. List all files in a directory and save the output to a text file.
2. Count the number of files in the directory.
3. Display the system's current date and time.

- Learning Outcome: Understanding basic Bash commands and script structure.

Assignment 2: Advanced Shell Scripting
- Task: Write a PowerShell script to:
  1. Monitor CPU and memory usage of a specific process.
  2. Log the results to a file every 10 seconds.
  3. Send an email alert if CPU usage exceeds 80%.
- Learning Outcome: Working with system monitoring and automation in Windows.

Assignment 3: Scripting for System Administration
- Task: Write a Bash script to:
  1. Create a backup of a directory by compressing it into a `.tar.gz` file.
  2. Schedule the backup script to run daily using `cron`.
- Learning Outcome: Automating repetitive tasks and using scheduling tools.

2. System Monitoring and Performance Tuning Assignments

Assignment 4: System Monitoring with `top` and `htop`
- Task:
  1. Use `top` to monitor CPU and memory usage of running processes.
  2. Identify the process consuming the most resources and terminate it.
  3. Repeat the task using `htop` and compare the two tools.
- Learning Outcome: Understanding system resource usage and process management.

Assignment 5: Performance Tuning with `vmstat`
- Task:
  1. Use `vmstat` to monitor system performance (CPU, memory, I/O).
  2. Identify bottlenecks and suggest optimizations (e.g., reducing running processes, increasing swap space).
- Learning Outcome: Analyzing system performance and tuning for efficiency.

Assignment 6: Windows Performance Monitoring with `perfmon` -
Task:
  1. Use `perfmon` to create a data collector set for monitoring CPU, memory, and disk usage.
  2. Analyze the collected data and generate a report.
- Learning Outcome: Using Windows tools for performance monitoring.

3. Process and Memory Management Assignments

Assignment 7: Simulating Process Scheduling Algorithms
- Task:
  1. Write a program (in C/Python) to simulate the following scheduling algorithms:
     - FCFS (First-Come, First-Served)
     - SJF (Shortest Job First)
     - Round Robin
  2. Compare the average waiting time and turnaround time for each algorithm.
- Learning Outcome: Understanding scheduling algorithms and their impact on system performance.

Assignment 8: Analyzing Memory Usage

- Task:
  1. Use tools like `free` and `vmstat` to monitor memory usage on a Linux system.
  2. Identify memory leaks in a sample program and fix them.
- Learning Outcome: Debugging memory-related issues and optimizing memory usage.

Assignment 9: Virtual Memory Simulation
- Task:
  1. Simulate demand paging and page replacement algorithms (FIFO, LRU, Optimal).
  2. Compare the number of page faults for each algorithm.
- Learning Outcome: Understanding virtual memory and page replacement strategies.

4. File System and Disk Management Assignments

Assignment 10: Creating and Managing File Systems
- Task:
  1. Create a new ext4 file system on a partition in Linux.
  2. Mount the file system and test read/write operations.
  3. Use `fsck` to check and repair the file system.
- Learning Outcome: Working with file systems and disk partitions.

Assignment 11: Simulating Disk Scheduling Algorithms
- Task:
  1. Write a program (in C/Python) to simulate the following disk scheduling algorithms:
     - FCFS (First-Come, First-Served)
     - SSTF (Shortest Seek Time First)
     - SCAN, C-SCAN
  2. Compare the total head movement for each algorithm.
  - Learning Outcome: Understanding disk scheduling and its impact on performance.

Assignment 12: File System Analysis
- Task:
  1. Use tools like `df`, `du`, and `lsblk` to analyze disk usage on a Linux system.
  2. Identify large files and directories and clean up unnecessary data.
- Learning Outcome: Managing disk space and optimizing storage.

5. Case Study Presentation

Assignment 13: OS Implementations Comparison
- Task:
  1. Form groups and prepare a presentation comparing Linux and Windows OS:
     - Architecture (kernel, file system, process management).
     - Performance (memory management, scheduling).
     - Security features (authentication, encryption).
  2. Include real-world use cases and examples.
  - Learning Outcome: Understanding the strengths and weaknesses of different OS implementations.

Suggested Textbooks and Resources

1. Textbooks:
   Operating System Concepts by Abraham Silberschatz, Peter B. Galvin, and Greg Gagne.
   Modern Operating Systems by Andrew S. Tanenbaum.

2. Online Resources:

Linux man pages and documentation.
Microsoft Windows documentation.
Online courses (e.g., Coursera, edX).

3. Tools:
VirtualBox/VMware for OS virtualization.
Linux terminal and Windows PowerShell.

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Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc. Computer Science-III (Major)

Title of the Course: : Internet Programming								
Year: III				**Semester: V**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-13	BS-CS352T	02	00	02	30	15	35	50

Learning Objectives:

- To Design dynamic and interactive Web pages.
- To Learn Core-PHP, Server Side Scripting Language
- To Learn PHP-Database handling

Course Outcomes (Cos)

On completion of the course, student will be able to–

- Understand how to develop dynamic and interactive Web Page

Detailed Syllabus:

Unit I: Overview of HTML, HTTP and PHP **(04)**

1.1 Overview of HTML and Basic Tags, Creating Forms, Tables, HTML5 Semantics.
1.2 CSS basic concept, three ways to use CSS, Box Model, Navigation Bar.
1.3 Introduction to Web server and Web browser.
1.4 HTTP basics.
1.5 PHP Basics: Use of PHP, Lexical structure, Language basics.
.

Unit II: Arrays **(06)**

2.1 Indexed Vs Associative arrays
2.2 Identifying elements of an array, Storing data in arrays, Multidimensional arrays Extracting multiple values
2.3 Converting between arrays and variables, Traversing arrays,
2.4 Sorting, Action on entire array

Unit III: Files and database handling **(10)**

3.1 Working with files and directories
3.2 Opening and Closing, getting information about file, Read/write to file, splitting name and path from file, Rename and delete files
3.3 Reading and writing characters in file Reading entire file
3.4 Random access to file data Getting information on file Ownership and permissions Using PHP to access a database Relational databases and SQL

3.5 PEAR DB basics
3.6 Advanced database techniques

Unit IV:Introduction to Web Techniques **(02)**

4.1Variables
4.2Server information
4.3Processing forms
4.4Setting response

Unit V:XML **(04)**

5.1What is XML?
5.2XML document
5.3StructurePHP and
5.4XML
5.5XML parser
5.6The document object
5.7modelThe simple XML
5.8Changing a value with simple XML

Unit VI:AJAX **(8)**

6.1Introduction of AJAX
6.2AJAX web
application
modelAJAX –PHP
framework
6.34Performing
AJAX validation
6.4.1Handling XML data using php
6.5 Connecting database using php and AJAX

Reference Books:

- HTML & CSS: The Complete Reference, Fifth Edition Author: Thomas A. Powell First published: 01 Jan 2010.
- Programming PHP by Rasmus Lerdorf and Kevin Tatroe, O'Reilly publication
- Beginning PHP 5, Wrox publication, PHP web services, Wrox publication
- Mastering PHP, BPB Publication, PHP cookbook, O'Reilly publication
- PHP for Beginners, SPD publication, Programming the World Wide Web, Robert W Sebesta (3rd Edition)
- HTML 5 Black Book : Covers Css3, JavaScript, XML, XHTML, Ajax, PHP And Jquery by Kogent Learning Solutions Inc., Published November 2011 by Dreamtech Press
- Spurlock Jake, Bootstrap: Responsive Web development. O'Reilly Media, Inc.

Ref. Links

- www.php.net.in
- www.W3schools.com
- www.wrox.com
- https://coreui.io/docs/layout/grid/#grid-options
  https://www.tutorialrepublic.com/twitter-bootstrap-tutorial/bootstrap-grid-system.php

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc. Computer Science-III (Major)

Title of the Course: Professional Practices in Python								
Year: III				**Semester: V**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-14	BS-CS353T	02	00	02	30	15	35	50

Learning Objectives:

- Student should be able to develop object oriented Programming logic using python
- Student should able to handle database using python
- Student should able to do network programming in python

Course Outcomes (Cos)

- Acquire knowledge and oops concept
- Students will learn the Advance concepts of python
- Student will learn Network programming and advance features in Python

Detailed Syllabus:

Unit I: Object Oriented Concept (8)

1.1 Python Classes and Object 1.1.1Constructor
1.1.2 init () method and self-argument
1.1.3 Attributes and instance variable
1.1.4 instance methods ()
1.1.3 Class variables
1.1.4 Class methods ()
1.1.4.1 Class method Decorator
1.1.4.2 Class method as Constructor
1.1.5 Static method
1.2 Encapsulation
1.3 Abstraction
1.4 Naming Convention
1.5 Name Mangling
1.6 Inheritance
1.6.1 Single Inheritance
1.6.2 Multiple Inheritance
1.6.3 Multilevel Inheritance

1.6.4 Hierarchical Inheritance
1.6.5 Hybrid Inheritance Polymorphism
1.7.1 Definition
1.7.2 inbuilt polymorphic functions
1.7.3 user-defined polymorphic functions
1.7.4 Polymorphism with class methods
1.7.5 Polymorphism with Inheritance
1.7.6 Polymorphism with a Function and objects
1.8 Abstract Class and interface
1.8.1 Abstract Method and Abstract Class
1.8.2 Interfaces in Python Abstract Classes vs. Interfaces
1.8.3

Unit II: Decorator, Iterator, Generator and comprehensions (6)

2.1 Decorator
2.1.1 function as object
2.1.2 Pass Function as Argument
2.1.3 Function returning a Function
2.1.4 introduction to python
2.1.3.1 Use of decorator
2.1.3.2 Syntax of decorators 2.1.3.3decorate a function with argument
2.1.3.3 decorator function returning a value
2.1.3.4 use of built-in decoder wrap
2.1.3.5 chaining decorators
2.1.3.6 decorator with argument
2.1.3.7 Memoization using decorators
2.2 Iterator
2.2.1 definition of Iterator
2.2.2 next () method
2.2.3 iter() method
2.3 Generator
2.3.1 yield keyword
2.3.2 Generator verses List
2.4 comprehensions
2.4.1 List comprehensions
2.4.2 nested list comprehensions
2.4.3 dictionary comprehensions
2.4.3 set comprehensions
2.4.4 Generator comprehensions

Unit III Multithreading (4)

3.1 Introduction
3.1.1 Use of thread
3.2 Thread modules
3.3 Methods form thread class
3.4 Function from thread class
3.5 Thread Local Data
3.6 Thread Objects in Python
3.7 Python Locks
3.8 Python RLocks and Semaphore Locks
Advantages and Disadvantages of Multithreading in Python

Unit IV: Database Connectivity **(8)**

4.1 Introduction

4.1.1 Generic database interface 4.1.2Relational database interface 4.1.3Benefits of python database programming

4.2 Advantages of a DBMS over files

4.3 Database Connection

4.4 Creating Cursor object

4.4 Creating table

4.5 Insert, read, update, delete operation execute & close functions reading single & multiple results of query execution executing different types of statements executing transactions

4.6 commit and rollback

4.7 Stored procedures

4.8 Disconnecting Database

4.9 Handling Errors understanding exceptions in database connectivity

Unit V: Network programming **(8)**

5.1 Introduction

5.2 Socket

5.3 Socket module

5.4 Server socket method

5.5 Client socket method

5.6 General socket method

5.7 Creating socket object

5.8 Function to setup the sever

5.9 Function to create client

5.10 creating server-client programs

5.11 sending email,

5.12 reading from URLPython internet modules

Suggested Readings/Material:

- Learning Python, 5th Edition
- Step By Step Database Programming using Python GUI & MySQL: A practical approach to learn database programming through progressive project-based examples

Foundations of Python Network Programming, Third Editio

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc. Computer Science -III (Major)

Title of the Course: Object Oriented Software Engineering								
Year: III				**Semester: V**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-15	BS-CS354T	00	02	02	30	15	35	50

Learning Objectives:

- Understanding importance of Object Orientation in Software engineering.
- Understand the components of Unified Modeling Language.
- Understand techniques and diagrams related to structural modeling.
- Understand techniques and diagrams related to behavioral modeling.

Course Outcomes (Cos)

- To Understand Object Oriented Modeling techniques and their applicability.
- To understand latest testing methods used in the software industries.

Detailed Syllabus:

Unit I: Object Oriented Concepts and Principles **(03)**

1.1 What is Object Orientation? -Introduction, Object, Classes and Instance, Polymorphism, Inheritance
1.2 Object Oriented System Development- Introduction, Function/Data Methods(With Visibility), Object Oriented Analysis, Object Oriented Construction.
1.3 Identifying the Elements of an Object Model.
1.4 Identifying Classes and Objects.
1.5 Specifying the Attributes (With Visibility).
1.6 Defining Operations.
1.7 Finalizing the Object Definition.

Unit II: Introduction to UML **(02)**

2.1 Concept of UML
2.2 Advantages of UML.

Unit III: Basic Structural Modeling **(04)**

3.1 Classes
3.2 Relationship
3.3 Common mechanism
3.4 Class Diagram (Minimum three examples should be covered)

Unit IV: Advanced Structural Modeling **(04)**

4.1 Advanced Classes

4.2 Advanced Relationship
4.3 Interface
4.4 Types and Roles
4.5 Packages
4.6 Object Diagram (Minimum three examples should be covered)

Unit V: Basic Behavioral Modeling (06)

3.1 Interactions
3.2 Use Cases and Use Case Diagram with stereo types (Minimum three examples should be covered)
3.3 Interaction Diagram (Minimum three examples should be covered)
3.4 Sequence Diagram (Minimum three examples should be covered)
3.5 Activity Diagram (Minimum three examples should be covered)
3.6 State Chart Diagram (Minimum three examples should be covered)

Unit VI: Object Oriented Analysis (05)

6.1 Interactive Development and the Rational Unified Process
6.2 Inception
6.3 Understanding Requirements
6.4 Use Case Model From Inception to Elaboration
6.5 Elaboration

Unit VII: Object Oriented Design (05)

7.1 The Booch Method, The Coad and Yourdon Method and Jacobson Method and Rambaugh Method
7.2 The Generic Components of the OO Design Model
7.3 The System Design Process – Partitioning the Analysis Model, Concurrency and Sub-System Allocation, Task Management Component, The Data Management Component, The Resource Management Component, Inter Sub System Communication
7.4 Object Design Process

Unit VIII: Architectural Modeling (06)

8.1 Component
8.2 Component Diagram (Minimum three examples should be covered)
8.3 Deployment Diagram (Minimum three examples should be covered)
8.4 Collaboration Diagram (Minimum three examples should be covered)

Reference Books:

1. Grady Booch, James Rambaugh, The Unified Modeling Language User/Reference Guide, Pearson Education INC
2. Ivar Jacobson, Object Oriented Software Engineering, Pearson Education INC
3. Craig Larman, Applying UML and Patterns Pearson Education INC Bennett, Simon, Object Oriented Analysis and Design McGraw Hill

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc. Computer Science -III (Major)

Title of the Course: Lab COURSE BS-CS352T and BS-CS353T								
Year: III				**Semester: V**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-16	BS-CS355P	00	02	02	30	15	35	50

Learning Objectives:

- To Design dynamic and interactive Web pages.
- To Learn Core-PHP, Server Side Scripting Language
- To Learn PHP- Database handling
- To apply statistical, data preprocessing and visualization techniques on data sets

Course Outcomes (Cos)

- Student understand the system calls in operating system & operations on processes
- Student understand CPU scheduling algorithm & Student understand how to handle deadlock
- Student understand the page replacement algorithm
- Understand how to develop dynamic and interactive Web Page
- Prepare data for use with a variety of statistical methods and recognize how the
- quality of the data may affect conclusions, Perform exploratory data analysis

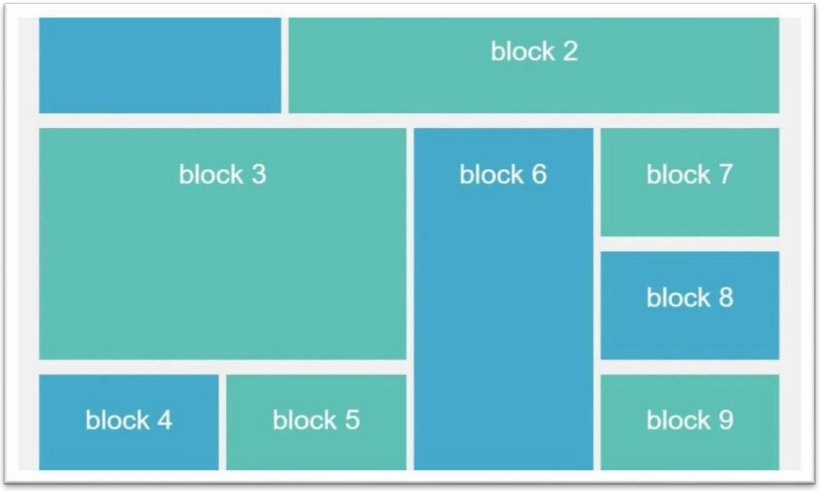
Detailed Syllabus:

Internet Programming – I

Practical Assignments

Assignment 1: Bootstrapping

1. Create following Bootstrap Web Layout Design. There are 9 blocks of the region in the arrangement. You can either place the images in them or the contents.



2. Create following BOotstrap Web Page Design

3. Create following layout using bootstrap

Assignment: 2 Using Arrays

1.Write a PHP script that inserts a new item in an array at any position. (hint : use

array_splice())

2.Write a menu driven program to perform the following stack and queue related operations:[Hint: use Array_push(), Array_pop(), Array_shift(), Array_unshift()]

a) Insert an element in stack
b) Delete an element from stack
c) Display the contents of stack
d) Insert an element in queue
e) Delete an element from queue
f) Display the contents of queue

3.Write a PHP script to sort the following associative array :
array("Sophia"=>"31","Jacob"=>"41","William"=>"39","Rames h"=>"40") in

a) ascending order sort by value
b) ascending order sort by Key
c) descending order sorting by Value
d) descending order sorting by Key

Assignment: 3 Using Files and DATABASE(PostgreSQL)

1. Write a program to read one file and display the contents of the file with its size.
2. Write a program to read a flat file "item.dat", which contains details of 5 different items such as Item code, Item Name, unit sold, and Rate. Display the Bill in tabular format
3.Property (pno, description, area)
   a. Owner (oname, address, phone)
   b. An owner can have one or more properties, but a property belongs to exactly one owner.
   c. Accept owner name from the user. Write a PHP script which will display all properties which are own by that owner

Assignment: 4 COOKIES and SESSIONS.

1. Write a PHP script to keep track of number of times the web page has been access. [Use session and cookies]

2. Write a PHP script to change the preferences of your web page like font style,font size, font color, background color using cookie. Display selected setting on next web page andactual implementation (with new settings) on third page.

3. Write a PHP script to accept username and password. If in the first threechances, username and password entered is correct then display second form with "Welcome message"otherwise display error message. [Use Session]

Assignment: 5 XML documents and DOM

1) Write a script to create XML file named "Item.xml"

```
<Item>
<ItemName>          </ItemName>
<ItemPrice>         </Itemprice>
<Quantity>          </Quantity>
</Item>
```

Link " Item. Xml" file to the CSS style sheet and get well formatted output asgiven below
i) ItemName :
Color : red;
Font-family : copperplate Gothic Light;Font-size : 16pt;
Font :bold;
ii) ItemPrice and Quantitycolor:yellow;

Python Assignments

Assignment 1 Understanding Object Oriented Programming, Python installation

Differentiate procedure oriented and Object Oriented Programming Identify a simple real world scenario using the concept of classes and objects

Demonstrate different types of inheritance in the scenario identified Practice Python Installation

Assignment 2: Develop the programs to implement parent-child relationship Demonstrate different inheritance types. Apply polymorphism and method overriding Create abstract classes

Assignment 3:

Implement the Python Classes and Objects to address the real world scenarios

Define classes and objects using python for the real world scenario Defining constructors and using Self Understand public and private members.

Practice calling class methods from another class Write built in functions to check, get, set and delete attributes.

Assignment 4:

Write the programs to address the exceptions via exception handling in the development of solutions and implement operator overloading.

Write a simple exception handling program with try- except Write a program for catching multiple exceptions Demonstrate raising and re raising exceptions

Apply else and finally clauses

Demonstrate the usage of polymorphism in overloading of operator

Assignment 5: Write program using Decorator

Assignment 6: Write program using iterator and generator

Assignment 7: Write program using comprehensions

Assignment 8: Write program on multitreading

Assignment 9: Write a program to demonstrate database connectivity in python

Assignment 10 : demonstrate network programming using python.

Write a program in Python to demonstrate following operations: a) Method overloading b) Method overriding

Assignment 10 : Write a program in Python to demonstrate following operations:

a) Simple inheritance b) Multiple inheritance

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc. Computer Science-III (Major)

Title of the Course: Core JAVA								
Year: III				**Semester: V**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSE-01	BS-CS356T(A)	02	00	02	30	15	35	50

Learning Objectives:

- To learn Object Oriented Programming language
- To study various java programming concept like Interface, File and Exception Handling etc.
- To design User Interface using Swing and AWT

Course Outcomes (Cos)

- To learn Object Oriented Programming language
- To study various java programming concept like Interface, File and Exception Handling etc.
- To design User Interface using Swing and AWT

.

Detailed Syllabus: Example

Unit I: An Introduction to Java (07)

Object Oriented Programming Concepts A short history of Java .
Features OR Buzzwords of Java
Java Environment Simple Java Program Java Tools – jdb, javap, javadoc
Types of Comments Data Types Final Variable Declaring 1D, 2D Array Accepting Input (Command Line Arguments, BufferedReader, Scanner)

Unit II: Objects and Classes (08)

Defining your own classes Access Specifiers (public, protected, private, default) Array of Objects Constructors, Overloading Constructors and Use of 'this' keyword static block, static fields And methods Predefined Classes
Object Class, Methods (equals(), toString(),hashcode(), getClass())
String Class And StringBuffer Class,Formatting String data using format() method
Creating , Accessing AndUsing Packages Wrapper Classes

Unit III: Inheritance and Interface (08)

Inheritance Basics (extends Keyword) and Types of Inheritance Superclass, Subclass and use of SuperKeyword Method Overriding and runtime polymorphism Use of final

keyword related to method and class Use of abstract class and abstract methods Defining and Implementing Interfaces Runtime polymorphism using interface Concept of Marker and Functional Interfaces.

Unit IV: Exception and File handling.

Dealing with errors , Exception class, Checked And Unchecked Exception Catching Exceptions, Multiple Catch Block, Nested try block Creating User Defined Exception Introduction to Files And Streams Input-OutputStream : FileInput/OutputStream, BufferedInput/OutputStream, DataInput/OutputStream Reader-Writer : FileReader/Writer, BufferedReader/Writer, InputStreamReader, OutputStreamWriter

Unit V: User Interface with AWT and Swing (10)

What is AWT? What is Swing? Difference between AWT and Swing The MVC ArchitectureAnd Swing Layouts And Layout Managers Containers And Components – JFrame, JButton, JLabel, JText, JTextArea, JCheckBox And JRadioButton, JList, JComboBox, JMenu And related Classes Dialogs (Message, Confirmation, Input), JFileChooser, JColorChooser Event Handling: Event Sources, Listeners Adapters And Anonymous Inner Class.

Suggested Readings:

1. Complete reference Java by Herbert Schildt(5th edition)
2. Java 2 programming black books, Steven Horlzner
3. Programming with Java , A primer ,Forth edition , By E. Balagurusamy

Core Java Volume-I-Fundamentals, Eighth Edition, Cay S. Horstmann, Gary Cornell, Prentice Hall,Sun Microsystems Pres

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc. Computer Science-III (Major)

Title of the Course: Data Analytics								
Year: III				**Semester: V**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSE-01	BS-CS356T(B)	02	00	02	30	15	35	50

Learning Objectives:

- Deploy the Data Analytics Lifecycle to address data analytics projects.
- Develop in depth understanding of the key technologies in data analytics.
- Apply appropriate analytic techniques and tools to analyze data, create models, and identify insights that can lead to actionable results.

Course Outcomes (Cos)

On completion of the course, student will be able to–

- Use appropriate models of analysis, assess the quality of input, and derive insight from results.
- Analyze data, choose relevant models and algorithms for respective applications
- Understand different data mining techniques like classification, prediction, clustering and association rule mining.
- Apply modeling and data analysis techniques to the solution of real world business problem.

Detailed Syllabus:

Unit I: Introduction to Data Analytics (06)

Concept of data analytics Data analysis vs Data analytics Types of analytics DiagnosticAnalytics, Predictive Analytics , Prescriptive Analytics, Exploratory Analysis, Mechanistic Analysis Mathematical models - Concept Model

evaluation: metrics for evaluating classifiers -Class imbalance - AUC, ROC (Receiver-Operator Characteristic)curves, Evaluating value prediction models

Unit II: Machine Learning Overview (06)

Introduction to Machine Learning, deep learning,
Artificial intelligence Applications for machine learning
in data scienceThe modeling process
Engineering features and selecting a model, Training the model, Validating themodel, Predicting new observations
Types of machine learning Supervised learning, Unsupervised learning,
Semi-supervisedlearning, ensemble techniques

Regression models Linear Regressions Polynomial Regression

Logistic Regressions Concept of classification, clustering and reinforcement learning.

Unit III: Mining Frequent Patterns, Associations, and Correlations (12)

What kind of patterns can be mined

Class/Concept Description: Characterization and Discrimination, Mining Frequent Patterns, Associations, and Correlations, Classification and Regressionfor PredictiveAnalysis, Cluster Analysis, Outlier Analysis

Mining frequent patterns - Market Basket Analysis.Frequent Itemsets, Closed Itemsets, and AssociationRules Frequent Item set Mining Methods, Apriori Algorithm

Generating Association Rules from FrequentItemsetsImproving efficiency of apriori algorithm,Frequent pattern growth (FP-growth) algorithm.

Unit IV: Social Media and Text Analytics (12)

Overview of social media analytics

Social Media Analytics Process, Seven layers of social media analytics, accessing Social media data Key social media analyticsmethodsSocial network analysis Link prediction, Community detection, Influence maximization, Expertfinding,Prediction of trust and distrust among individuals

Introduction to Natural Language Processing

Text Analytics : Tokenization, Bag of words, Word weighting : TF-IDF, n-Grams, stopwords,Stemming and lemmatization, synonyms and parts of speech tagging

Sentiment Analysis Document or text summarization Trend analytics

Challenges to social media analytics

Suggested Readings:

1) Data Science Fundamentals and Practical Approaches, Gypsy Nandi, Rupam Sharma, BPBPublications,2020.
2) The Data Science Handbook, Field Cady, John Wiley & Sons, Inc, 2017
3) Data Mining Concepts and Techniques, Jiawei Han, Micheline Kamber, Jian Pei, MorganKaufmann, ThirdEdition, 2012.
4) A Hands-On Introduction to Data Science, Chirag Shah, University of WashingtonCambridge UniversityPress
5) The Data Science Design Manual, Steven S. Skiena, Springer, 2017
Introducing data science: big data, machine learning, and more, using Python tools

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar (Autonomous)

Syllabus

B.Sc. Computer Science-III (Major)

Title of the Course: Lab Course based on BS-CS356T(A)								
Year: III				**Semester: V**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSE-02	BS-CS357P (A)	02	00	02	30	15	35	50

Learning Objectives:

- Covers the complete scope of the syllabus.
- Bringing uniformity in the way course is conducted across different colleges.
- Continuous assessment of the students.

Course Outcomes (Cos)

- Use an integrated development environment to write, compile, run,
- and test simple object-oriented Java programs.
- Read and make elementary modifications to Java programs that solve real-world problems.
- Validate input in a Java program.

Detailed Syllabus: Example

Assignment 1 :
Java Tools and IDE, Simple java programs [Slot – 1]
Introduction to the java environment Use of java tools like java, javac, jdb and javadoc Defining simpleclasses and creating objects.

Assignment 2 :
Array of Objects and Packages [Slot – 2] Defining a class. Creating an array of objects. Creating a package.

Assignment 3 :
Inheritance and Interfaces [Slot – 2] To implement inheritance in java. To define abstract classes. Todefine and use interfaces and Functional Interface.

Assignment 4 :
Exception and File Handling [Slot – 2] Demonstrate Exception Handling Mechanism in Java. Use of try, catch, throw, throws, finally blocks Defining User defined Exception classes. Creation of files and demonstration of I-O operations

Assignment 5 :
GUI Designing, Event Handling [Slot – 5] To demonstrate GUI creation using Swing Package and Layout managers. To understand Event handling mechanism in Java. Using Event classes, Event Listeners and Adapters

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar (Autonomous)

Syllabus

B.Sc. Computer Science-III (Major)

Title of the Course: Lab Course based on BS-CS356T(B)								
Year: III				**Semester: V**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSE-02	BS-CS357P(B)	02	00	02	30	15	35	50

Learning Objectives:

- To understand the impact of finding frequent patterns from large datasets.
- To learn the Apriori Algorithm which is used for frequent itemsets mining.
- To understand Association Rule Mining.
- To write and learn implementation of such concepts with Python.
- Apply appropriate analytic techniques and tools to analyze data, create models, and identify insights that can lead to actionable results.
- Apply modeling and data analysis linear and logistic regression techniques to the solution of real world business problems.

Course Outcomes (Cos)

- Data Analytics application such as structured statistical and mathematical techniques are studied.
- Understand the key issues in big data management and its associated applications in intelligent business and scientific computing.
- Able to analyse business data using regression model.

Detailed Syllabus:

Sr. No.	**Assignment Title**
1	Linear and Logistic Regression
2	Frequent itemset and Association rule mining
3	Text and Social Media Analytics

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar (Autonomous)

Syllabus

B.Sc. Computer Science-III (Major)

Title of the Course: Theoretical computer Science								
Year: III				**Semester: V**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
VSC-03	BS-CS358T	02	00	02	30	15	35	50

Learning Objectives:

- To understand the Finite Automata, Pushdown Automata and Turing Machine.
- To understand the Regular Language, Context Free Language, Context Sensitive Language and Unrestricted Language.
- To understand the relation between Automaton and Language

Course Outcomes (Cos)

On completion of the course, student will be able to–

- Understand the use of automata during language design.
- Relate various automata and Languages.

Detailed Syllabus:

Unit I: Finite Automaton (10)

1.1 Introduction: Symbol, Alphabet, String, Prefix & Suffix of Strings, Formal Language, Operations on Languages.

1.2 Deterministic finite Automaton – Definition, DFA as language recognizer,DFA as pattern recognizer

1.3 Nondeterministic finite automaton – Definition and Examples. NFA To DFA (Myhill Nerode Method) NFA with ε- transitions Definition and Examples.NFA with εTransitions to DFA & Examples

1.4 Finite automaton with output – Mealy and Moore machine, Definition and Examples.

1.5Minimization of DFA, Algorithm & Problem using Table Method

Unit II: Regular Expressions and Languages (06)

2.1 Regular Expressions (RE): Definition & Example Regular Expressions Identities.

2.2Regular Language-Definition and Examples.
Conversion of RE to FA-Examples.

2.3 Pumping lemma for regular languages and applications. Closure Properties of regular Languages

Unit III: Context-Free Grammars and Languages **(10)**

3.1 Grammar - Definition and Examples.
3.2 Derivation-Reduction - Definition and Examples .Chomsky Hierarchy.
3.3 CFG: Definition & Examples. LMD, RMD, Parse Tree Ambiguous
3.4 Grammar: Concept & Examples.
3.5 Simplification of CFG: Removing Useless Symbols, Unit Production, ϵ-production and Nullable Symbol. 3.6 Normal Forms: Greibach Normal Form (GNF) and Chomsky Normal Form (CNF)Regular Grammar: Definition.
3.7 Left linear and Right Linear Grammar-Definition and Example.Equivalence of FA & Regular Grammar
3.8 Construction of regular grammar equivalent to a given DFA.
3.9 Construction of a FA from the given right linear grammar

Unit IV: Push Down Automata (05)

4.1 Definition of PDA and examples.
4.2 Construction of PDA using empty stack and final State method: Examples using stackmethod.
4.3 Definition DPDA & NPDA, their correlation and Examples of NPDA CFG (in GNF) to PDA: Method and examples

Unit V: Turing Machine (05)

5.1 The Turing Machine Model, Definition and Design of TMProblems on language recognizers. Language accepted by TM.
5.2 Types of Turing Machines (Multitrack TM, Two-way TM, Multitape TM, Nondeterministic TM)
5.3 Introduction to LBA (Basic Model) & CSG. (Without Problems)

Suggested Readings/Material:

- Introduction to Automata Theory, Languages and Computation, John E. Hopcraft, Rajeev Motwani, Jeffrey D. Ullman, Third Edition, Pearson Education Publication, 2008
- Introduction to Automata theory, Languages and computation By John E. Hopcroft and JeffreyUllman – Narosa Publishing House, 1995
- Theory of Computer Science Automata, Languages and Computation, K.L.P. Mishra, N. Chandrasekaran, Publication- Prentice Hall of India, 2008
- Introduction to Computer Theory Daniel I. A. Cohen – 2 nd edition – John Wiley & Sons,1996
- Introduction to Languages and The Theory of Computation John C. Martin The McGraw-Hill, Fourth Edition, 2011

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc. Computer Science-III (Major)

Title of the Course: Field Project								
Year: III				**Semester: V**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
FP-02	BS-CS359P	00	02	02	60	15	35	50

1. The department shall conduct the awareness programme/session on how to complete the Field Project (Major related) at the beginning of the semester.
2. The department forms batches and appoints guides/mentors for each batch.
3. The department includes a session of 04 hours/batch in the timetable to guide and evaluate students every week.
4. Students need to prepare a detailed field project report and submit it to the department.
5. The Field Project will be evaluated based on presentation and report as per guidelines.
6. Field Project Guidelines Approved by Academic Council:
   https://newartscollege.ac.in/wp-content/uploads/2024/07/Field_Project_Guidelines1.pdf

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar (Autonomous)

Syllabus

B.Sc.(Computer Science)-III (Major)

Title of the Course: React JS								
Year: III				**Semester: VI**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-17	BS-CS360T	02	00	02	30	15	35	50

Learning Objectives:

- Create and render React components using JSX syntax.
- Work with React components, both class and functional components.
- Handle component props and state management.
- Implement event handling and conditional rendering.
- Create and manage lists with keys in React.
- Build forms and handle user input in React applications.
- Implement navigation using React Router.

Course Outcomes (Cos)

- To build user interfaces specifically for single-page applications.
- To enable fast and quality assured application development that in turn saves time for both - clients and developers.
- To create reusable UI components and hence reduce the code.

Course Contents:

Unit I: Introduction to React JS **(04)**

1.1 What is ReactJS?
1.2 History of ReactJS.
1.3 Why to learn ReactJS?
1.4 Features of ReactJS.
1.5 Advantages of ReactJS.
1.6 Aspects of ReactJS.

Unit II: ReactJS Installation **(06)**

2.1 Setting up Environment for ReactJS
2.2 NodeJS Installation
2.3 Downloading and adding ReactJS Libraries.
2.4 Creating your first ReactJS project.
2.5 Use of create-react-app
2.6 ReactJS Project Structure

Unit III: ReactJS ES6 **(06)**

3.1 Classes
3.2 Arrow Functions
3.3 Variables
3.4 Array Methods
3.5 DeStructuring
3.6 Modules
3.7 Ternary Operators and Spread Operators.

Unit IV: React JS Fundamentals **(08)**

4.1 React JSX
4.2 React Component
4.2.1 Function Based Component
4.2.2 Class Based Component
4.3 React Component API
4.4 Component Lifecycle
4.5 React State
4.6 React Props
4.7 React State vs React Props

Unit V: Form Handling in ReactJS **(06)**

5.1 React Forms
5.2 React Events
5.3 React Lists
5.4 React Keys
5.5 React Refs
5.6 React Fragments

Suggested Readings:

1. "Full Stack React" The Complete Guide to ReactJS and Friends

2. The Road to Learn React: Your journey to master plain yet pragmatic React.js byRobinWieruch
3. React in Action by Mark Tielens Thomas
4. Learning React, 2nd Editi

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar (Autonomous)

Syllabus

B.Sc.(Computer Science)-III (Major)

Title of the Course: Compiler Construction								
Year: III				**Semester: VI**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-18	BS-CS361T	02	00	02	30	15	35	50

Learning Objectives:

- To understand design issues of a lexical analyzer and use of LEX tool.
- To understand design issues of a parser and use of YACC tool.
- To understand and design code generation and optimization techniques.

Course Outcomes (Cos):

On completion of the course, student will be able to–

- Understand the process of scanning and parsing of source code.
- Learn the conversion code written in source language to machine language.
- Understand tools like LEX and YACC.

Course Contents:

Unit I: Introduction (04)

1.1 Definition of Compiler, Aspects of compilation.
1.2 The structure of Compiler.
1.3 Phases of Compiler – Lexical Analysis, Syntax Analysis, Semantic Analysis, Intermediate Code generation, code optimization, code generation, Error Handling.
1.4 Introduction to one pass & Multi pass compilers, cross compiler, Bootstrapping.

Unit II: Lexical Analysis (Scanner) (04)

2.1 Review of Finite automata as a lexical analyzer,
2.2 Applications of Regular Expressions and Finite Automata (lexical analyzer, searching using RE), Input buffering, Recognition of tokens.
2.3 LEX: A Lexical analyzer generator (Simple Lex Program)

Unit III: Syntax Analysis (Parser) (14)

3.1 Definition, Types of Parsers
3.2 Top-Down Parser –
3.2.1 Top-Down Parsing with Backtracking: Method & Problems
3.2.2 Drawbacks of Top-Down parsing with backtracking,
3.2.3 Elimination of Left Recursion (direct & indirect) 3.2.4Need for Left Factoring & examples
3.3 Recursive Descent Parsing: Definition

3.3.1 Implementation of Recursive Descent Parser Using Recursive Procedures
3.4 Predictive [LL (1)] Parser (Definition, Model)
3.4.1 Implementation of Predictive Parser [LL (1)]
3.4.2 FIRST & FOLLOW
3.5 Construction of LL (1) Parsing Table
3.5.1 Parsing of a String using LL (1) Table.
3.6 Bottom-Up Parsers
3.7 Operator Precedence Parser -Basic Concepts
3.7.1 Operator Precedence Relations Form Associativity & Precedence Operator Precedence Grammar
3.7.2 Algorithm for LEADING & TRAILING (with ex.)
3.7.3 Algorithm for Operator Precedence Parsing (with ex) Precedence Functions
3.8 Shift Reduce Parser
3.8.1 Reduction, Handle, Handle Pruning
3.8.2 Stack Implementation of Shift Reduce Parser (with examples)
3.9 LR Parser: Model, Types [SLR (1), Canonical LR, LALR]-Method & examples.
3.10 YACC (from Book 3) –program sections, simple YACC program for expression evaluation

Unit IV: Syntax Directed Definition (07)

4.1 Syntax Directed Definitions (SDD) Inherited & Synthesized Attributes
4.2 Evaluating an SDD at the nodes of a Parse Tree, Example Evaluation Orders for SDD's Dependency Graph.
4.3 Ordering the Evaluation of Attributes S-Attributed Definition, L-Attributed Definition. Application of SDT Construction of syntax trees,
4.4 Translation Schemes.
4.4.1 Definition, Postfix Translation Scheme

Unit V: Code Generation and Optimization (07)

5.1 Compilation of expression.
5.1.1 Concepts of operand descriptors and register descriptors with example. Intermediate code for expressions – postfix notations, Triples, Quadruples and Expression trees.
5.2 Code Optimization – Optimizing transformations – compile time evaluation, elimination of common sub expressions, dead code elimination, frequency reduction, strength reduction.
5.2.1 DAG for Three address code.
5.2.2 The Value-number method for constructing DAG's.
5.2.3 Definition of basic block, Basic blocks, and flow graphs Directed acyclic graph (DAG) representation of basic block. Issues in design of code generator

Suggested Readings/Material:

- Compilers: Principles, Techniques, and Tools, Alfred V. Aho, Ravi Sethi, Jeffrey D. Ullman, 2004.
- Principles of Compiler Design By: Alfred V. Aho, Jeffrey D. Ullman, Narosa PublicationHouse, 2002 LEX & YACC, 2nd edition, O'reilly Publication, 2012.

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc.(Computer Science)-III (Major)

Title of the Course: Android Programming								
Year: III				**Semester: VI**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-19	BS-CS362T	02	00	02	30	15	35	50

Learning Objectives:

- Develop competence and confidence in android programming.
- Understand the entire Android Apps Development Cycle.
- Enable the students to independently create Android Applications.

Course Outcomes (Cos):

After successful completion of this course, learner will be able to:

- Describe the process of developing mobile applications.
- Create mobile applications on the Android Platform.
- Design and implement mobile applications involving data storage in SQLite database
- Use location-based services while developing applications

Course Contents:

Unit I: Introduction to Android **(05)**

1.1. Overview
1.2. History
1.3. Features of Android
1.4. Architecture of Android
• Overview of Stack • Linux Kernel • Native Libraries • Android Runtime.
• Application Framework • Applications.
1.5. SDK Overview
• Platforms.
• Tools – (JDK, SDK, Eclipse/Android Studio, ADT, AVD, Android Emulator), Versions.
1.6. Creating your first Android Application.

Unit II: Activities, Fragments and Intents **(07)**

2.1. Introduction to Activities.
2.2. Activity Lifecycle.
2.3. Introduction to Intents.
2.4. Linking Activities using Intents.
2.5. Calling built-in applications using Intents.
2.6. Introduction to Fragments.
2.7. Adding Fragments Dynamically.
2.8. Lifecycle of Fragment.

2.9. Toast.

Unit III: Android User Interface (07)

3.1. Understanding the components of a screen

• Views and View Groups • Linear Layout • Absolute Layout • Table Layout.

• Relative Layout • Frame Layout • Scroll Layout • Scroll View • Constraint Layout.

3.2. Adapting to Display Orientation

• Anchoring Views.

• Resizing and Repositioning.

3.3. Split Screen / Multi-Screen Activities.

Unit IV: Designing Your User Interface with Views (10)

4.1. Using Basic Views

• TextView.

• Button, ImageButton, EditText, CheckBox.

• Switch, ToggleButton, RadioButton, and RadioGroup Views.

• ProgressBar View.

• AutoCompleteTextView View.

4.2. Using Picker Views

• TimePicker View • DatePicker View.

4.3. Using List Views to Display Long Lists

• ListView View • Using the Spinner View.

4.4. Understanding Specialized Fragments

• Using a ListFragment • Using a DialogFragment.

4.5. Displaying Pictures and Menus

Using Image Views to Display Pictures.

• Gallery and ImageView views.

• Image Switcher.

• Grid View.

Using Menus with Views.

• Creating the helper methods • Options Menu • Context Menu.

4.6. VideoView.

• Play video from URL with using VideoView.

• VideoView Create.

• Optimized VideoView.

• Optimized VideoView in ListView.

Unit V: Databases – SQLite. (05)

5.1. Introduction to SQLite.

5.2. SQLite Open Helper and SQLite Database.

5.3. Creating, opening and closing database.

5.4. Working with cursors, Insert, Update, Delete.

5.5. Building and executing queries.

Suggested Readings/Material:

- Beginning Android4 Application Development, By Wei-Meng Lee WILEY India Edition WROX Publication.
- Professional Android 4 Application Development, By Reto Meier WROX Publication.
- Head First Android Development: A Brain-Friendly Guide, By David Griffiths and Dawn Griffiths.

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc.(Computer Science)-III (Major)

Title of the Course: Lab Course Based on BS- CS360T								
Year: III				**Semester: VI**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSC-20	BS-CS363P	00	02	02	30	15	35	50

Learning Objectives:

- How to set up a React development environment and create a new React project
- Understand the core concepts of React, such as components, state, and props
- Implement common React patterns and best practices
- Build various React components, including functional and class-based components
- Manage state and data flow in React applications

Course Outcomes (Cos):

On completion of the course, student will be able to–

- To build user interfaces specifically for single-page applications.
- To enable fast and quality assured application development that in turn saves time for both - clients and developers.
- To create reusable UI components and hence reduce the code.

Course Contents:

List of Assignments:

1. Setup the environment for ReactJS.
   a. Install NodeJS.
   b. Download ReactJS Libraries.
   c. Create ReactJS App using create-react-app.
2. Create a simple ReactJS app and display "Hello World" message on webpage.
3. Create a list of programming languages you know and display the list onwebpage.
4. Create a class-based Component Car and display the car details.
5. Create a function-based Component and display your own details like Name,DOB, Contact Number, Address etc.
6. Add am attribute to pass a colour to the Car Component and use itrender()function to display the Colour of Car.
7. Create a class Component Car and display Car Component in root.
8. Create a Constructor function in the Car Component and add a Colour property.Use the Colour property in render function to display it.
9. Create a class Component called as Garage. Use the Car Component inside theGarage Component.
10. Specify the state object in the constructor method. The state object can contain as many properties as you like. Add all properties of your CarComponent like brand, model,

colour, year, etc.

11. Refer to the state object and display Car details using render() function.
12. Add a button with an onClick() event that will change the colour property.
13. Implement lifecycle methods of Component.
14. Add a Brand attribute to the Car element. Use the props object to accessbrandattribute in Component.
15. Send this Brand property from Garage Component to Car Component.
16. Create a Football Component. Put a shoot() function in it. Call this shoot()on abutton click.
17. Create a list of Car brands in Garage Component. Render all the Car fromGarage.
18. Add keys to above list items of Brands in Garage Component. Render all theCarfrom Garage.
19. Add an attribute that allows React to keep track of elements in list.
20. Create a form that allow user to enter their name. Use onChange() to manage the input.
21. Add a submit button and an event handler in the onSubmit attribute.

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc.(Computer Science)-III (Major)

Title of the Course: Advanced JAVA								
Year: III				**Semester: VI**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSE-03	BS-CS364T(A)	02	00	02	30	15	35	50

Learning Objectives:

- To learn database programming using Java
- To study web development concept using Servlet and JSP
- To develop a game application using multithreading
- To learn socket programming concept

Course Outcomes (Cos):

On completion of the course, student will be able to–

- To access open database through Java programs using Java Data Base Connectivity (JDBC) and develop the application.
- Understand and Create dynamic web pages, using Servlets and JSP.
- Work with basics of framework to develop secure web applications.

Course Contents:

Unit I: Collections (06)

Introduction to the Collection framework, List - ArrayList, LinkedList Set - HashSet, TreeSet, Map -HashMap and TreeMap, Interfaces such as Comparator, Iterator, ListIterator, Enumeration.

Unit II: Multithreading (06)

What are threads? Life cycle of thread Creating threads - Thread class , Runnable interface ,Thread priorities, Running multiple threads Synchronization and interthread communication

Unit III: Database Programming (06)

The design of jdbc Types of drivers Executing sql statements, query execution Scrollableand updatable Resultset

Unit IV: Servlets and JSP (12)

Introduction to Servlet and Hierarchy of Servlet, Life cycle of servlet ,Handing get and post request, (HTTP) Handling data from HTML to servlet ,Retrieving data from database to servlet, Session tracking – User Authorization, URL rewriting, Hidden form fields, Cookies and HttpSession Introduction to JSP, Life cycle of JSP ,Implicit Objects Scripting elements Declarations, Expressions, Scriplets, Comments JSP Directives - Page Directive, include directive Mixing Scriplets and HTML JSP Actions - jsp:forward, jsp:include, jsp:useBean, jsp:setProperty and jsp:getProperty

Unit V: Spring Framework **(06)**

Introduction of Spring framework Spring Modules / Architecture, Spring Applications Spring MVCSpring MVC Forms, Validation.

Suggested Readings/Material:

- Complete reference Java by Herbert Schildt(5th edition)
- Java 2 programming black books, Steven Horlzner
- Programming with Java , A primer ,Forth edition , By E. Balagurusamy
- Core Java Volume-I-Fundamentals, Eighth Edition, Cay S. Horstmann, Gary Cornell, Prentice Hall,Sun Microsystems Press
- Core Java Volume-II-Advanced Features, Eighth Edition, Cay S. Horstmann, Gary Cornell, PrenticeHall, Sun Microsystems Press
- Getting started with Spring Framework: covers Spring 5 by J Sharma and Ashish Sarin
- Spring 4 for Developing Enterprise Applications: An End-to-End Approach by Henry H. Liu.

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc.(Computer Science)-III (Major)

Title of the Course: Soft Computing								
Year: III				**Semester: VI**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSE-03	BS-CS364T(B)	02	00	02	30	15	35	50

Learning Objectives:

- To learn the basic concepts of Soft Computing
- To become familiar with various techniques like neural networks, genetic algorithms and fuzzy systems.
- To apply soft computing techniques to solve problems.

Course Outcomes (Cos):

- To introduce the ideas of soft computational techniques based on human experience.
- To generate an ability to design, analyze and perform experiments on real life problems using various Neural Learning Algorithms.
- To conceptualize fuzzy logic and its implementation for various real world applications.
- To apply the process of approximate reasoning using Neuro- Fuzzy Modeling.
- To providethe mathematical background to carry out optimization using genetic algorithms.

Course Contents:

Unit I: Introduction to Soft Computing (02)

Neural Networks: Definition, Advantages, Applications, Scope.
Fuzzy logic: Definition, Applications. Genetic Algorithms: Definition, Applications

Unit II: Neural Network

Fundamental Concept: Artificial Neural Network, Biological Neural Network, Brain vs. Computer-Comparison Between Biological Neuron and Artificial Neuron (Brain vs. Computer), Artificial Neurons, Neural Networks and Architectures: Neuron Abstraction, Neuron Single Functions, Mathematical Preliminaries, Neural Networks Defined, Architectures: Feedforward and Feedback, Salient Properties of Neural Networks Geometry of Binary Threshold Neurons and Their Networks: Pattern Recognition and Data Classification, Convex Sets, Convex Hulls and Linear Separability, Space of Boolean Functions, Binary Neurons are Pattern Dichotomizers,Non-linearly Separable Problems, Capacity of a Simple Threshold Logic Neuron, Revisiting the XOR Problem, Multilayer Networks, How Many Hidden Nodes areEnough?

Learning and Memory: An Anecodatal Introduction, Long Term Memory, The Behavioral Approach to Learning, The Molecular Problem of Memory, Learning Algorithms, Error

Correction and Gradient Descent Rules, Learning Objective for TLNs, Pattern Space and Weight Space. Linear Seperabilty, Hebb Network, Perceptron Network. α- Least Mean Square Learning, MSE Error Surface and Its Geometry, Steepest Descent Search with Exact Gradient Information, μ-LMS: Approximate Gradient Descent,
Application of LMS TO Noise Cancelation.

Unit III: Fuzzy Set Theory

Brief Review of Conventional Set Theory, Introduction to Fuzzy Sets, Properties of Fuzzy Sets, Operations on Fuzzy Sets, Crisp Relation, Fuzzy Relation, Tolerance and equivalence relation, Fuzzy Tolerance and equivalence relation, Fuzzy Max-Min and Max-Product Composition, Membership Functions, Fuzzification, Defuzzification to crisp sets, λ-Cuts for fuzzy Relations, Fuzzy (Ruled-Based) system, Graphical technique of inference, Membership value assignment-Intuition, Inference. Fuzzy Classification -Classification by equivalence relation, Cluster analysis, cluster validity, c-Means clustering, Hard c-means, Fuzzy c-Means, Fuzzy Arithmetic,
Fuzzy Extension Principle.

Unit IV: Genetic Algorithms

What are Genetic Algorithms? Why Genetic Algorithms? Biological Background: The Cell, Chromosomes, Genetics, Reproduction, Natural Selection, Traditional Optimization and Search Techniques, Simple GA, Terminologies and Operators in GA, Encoding, Selection, Crossover, Mutation, Search Termination, Constraints in GA,
Problem solving using GA, Classification of GA.

Suggested Readings/Material:

Sr. No.	Title of the Book	Author/s	Publication
1	Fuzzy Logic With Engineering Applications	Timothy Ross	Wiley Publication
2	Introduction to Soft Computing	Deepa & Shivanandan	Wiley Publication
3	Genetic Algorithms in Search, Optimization and Machine Learning	David E. Goldberg	Pearson Education
4	Fundamentals of Neural Networks – Architectures, Algorithms, And Applications	Laurene Fausett	Pearson Education
5	Neural Networks	Satish Kumar	Tata McGrawHill

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar (Autonomous)

Syllabus

B.Sc.(Computer Science)-III (Major)

Title of the Course: Practical Course Based on CS 364T(A)								
Year: III				**Semester: VI**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSE-04	BS-CS365P	00	02	02	30	15	35	50

Learning Objectives:

- Covers the complete scope of the syllabus.
- Bringing uniformity in the way course is conducted across different colleges.
- Continuous assessment of the students.
- Advanced Java is designed to develop web based, network centric, Enterprise level applications

Course Outcomes (Cos):

- To Learn Database Programming using Java
- Understand and Create dynamic web pages using Servlets and JSP.
- Work with basics of framework to develop secure web applications

Course Contents:

Guidelines: Operating Environment:

- Operating system: Linux Editor: Anylinux based editor like vi, gedit and Use of IDE – Eclipse etc.
- Compiler: javac
- Database: postgresql.

List of Assignments:

Assignment 1:

Database Programming [Slot-2] Study the Collection framework in java. To Implement Various Interfaces and classes through algorithms. To Demonstrate Cursor Objects (Enumeration, Iterator,ListIterator, Comparator).

Assignment 2:

Multithreading [Slot-2] To create and use threads in java. To demonstrate multithreading using Thread Synchronization, Inter-thread. Communication, Thread Priorities.

Assignment 3:

Database Programming [Slot-2] To communicate with a database using java. To execute queries on tables. To obtain information about the database and tables.

Assignment 4:

Servlets [Slot-2] To understand server-side programming. Simple steps to create and execute servlets. How to pass parameters using doGet and doPost methods? Handling data from HTML to servlet. How to connect servlet to a database? Use of various session tracking methods like Cookies.

Assignment 5:

Java Server Pages [Slot-2] JSP life-cycle. Use of JSP implicit objects. JSP Directives. Use of ScriptingElements. To understand actionstags in JSP. Understanding flow of JSP custom tags.

Assignment 6:

Spring Framework [Slot-2] To create and understand the steps to develop Spring application

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc.(Computer Science)-III (Major)

Title of the Course: Practical Course Based on CS 364T(B)								
Year: III				**Semester: VI**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
DSE-04	BS-CS365P	00	02	02	30	15	35	50

Learning Objectives:

After completing this course, students will be able to learn:

- Fuzzy logic and its applications.
- Artificial neural networks and its applications.
- Solving single-objective optimization problems using GAs.
- Solving multi-objective optimization problems using Evolutionary algorithms (MOEAs).
- Applications of Soft computing to solve problems in varieties of application domains.

Course Outcomes (Cos):

- Understand appropriate learning rules for each of the architectures and learn several neural network paradigms and its applications**.**
- Students Recognize the feasibility of applying a soft computing methodology for a particular problem.
- Students Reveal different applications of these models to solve engineering and other problems

Course Contents:

Implement the programs in C/C++/Java/MATLAB:

Assignments

1. Write a program to implement Fuzzy Operations: Union, Intersection, Complement, algebraic sum, Algebraic product, Cartesian product.
2. Write a program to implement De Morgans law.
3. Write a program to implement Max-Min Composition and Max-Product Composition.
4. Write a program to implement lambda cut.
5. Write a program to implement Activation Function.
6. Write a program to implement Perceptron Learning Rule.
7. Write a program to implement Hebb's Rule.
8. Write a program to implement Feed Forward Network.

9. Write a program for building an Artificial Neural Network by implementing the Back Propagation Algorithm and test the same using appropriate data sets.
10. Write a program for solving linearly separable problem using Perceptron Model.
11. Write a program to develop supervised learning algorithm.
12. Write a program to study and analyze genetic life cycle.

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc.(Computer Science)-III (Major)

Title of the Course: Practical Course Based on 362T								
Year: III				**Semester: VI**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
VSC-04	BS-CS366P	00	02	02	60	15	35	50

Learning Objectives:

- To be able to understand the process of developing software for the mobile.
- To be able to create mobile applications on the Android Platform.
- To be able to create mobile applications involving data storage in SQLite database.

Course Outcomes (Cos):

After successful completion of this course, learner will be able to:

- Describe the process of developing mobile applications.
- Create mobile applications on the Android Platform.
- Design and implement mobile applications involving data storage in SQLite database.
- Use location-based services while developing applications.

Course Contents:

Assignment No 1. Activities, Fragments and Intents.

Assignment No 2. Android User Interface.

Assignment No 3. Designing Your User Interface with Views.

Assignment No 4. Databases – SQLite.

Ahmednagar Jilha Maratha Vidya Prasarak Samaj's

New Arts, Commerce and Science College, Ahmednagar
(Autonomous)
Syllabus
B.Sc.(Computer Science)-III (Major)

Title of the Course: On Job Training								
Year: III				**Semester: VI**				
Course Type	Course Code	Credit Distribution		Credits	Allotted Hours	Allotted Marks		
		Theory	Practical			CIE	ESE	Total
OJT-01	BS-CS367P	00	04	02	60	15	35	50

Course Outcomes (Cos)

1. To explore the subject-related professional competency through OJT/Internship
2. To learn professional skills and practices in Government or Non-government or Research Organizations, Industry, subject-related establishments etc.
3. To develop a network with industry personnel/experts or authorities to build a career after graduation.

Duration:

The OJT/internship is of 120 hours. The students need to complete the selected OJT/internship for the fulfilment of the mandatory 04 credits in the VI semester of graduation.

Note: Students can complete the OJT/internship/s during the II semester condition to compulsorily attending the theory and practical sessions of the other courses prescribed in the semester or after the conclusion of the VI semester examination during the summer vacations (Summer Term).

The last date of submission of the OJT/Internship report will be on or before the 30th day after the conclusion of the II semester examination every year or the date declared by the college/department in the specific academic year.

Detail Guidelines and Report Formats are available on the following link:

1. Internship Guidelines:

   https://newartscollege.ac.in/wp-content/uploads/2024/07/Internship_Guidelines1.pdf

2. OJT Guidelines for III Year UG:

   https://newartscollege.ac.in/wp-content/uploads/2024/07/OJT-Guidlines_0001.pdf